The Invariance Principle — problem 1.58

A. Engel — Problem-Solving Strategies · 1998

Consider any binary word $W = a_1a_2\cdots a_n$. It can be transformed by inserting, deleting or appending any word XXX, X being any binary word. Our goal is to transform W from 01 to 10 by a sequence of such transformations. Can the goal be attained (LMO 1988, oral round) ?